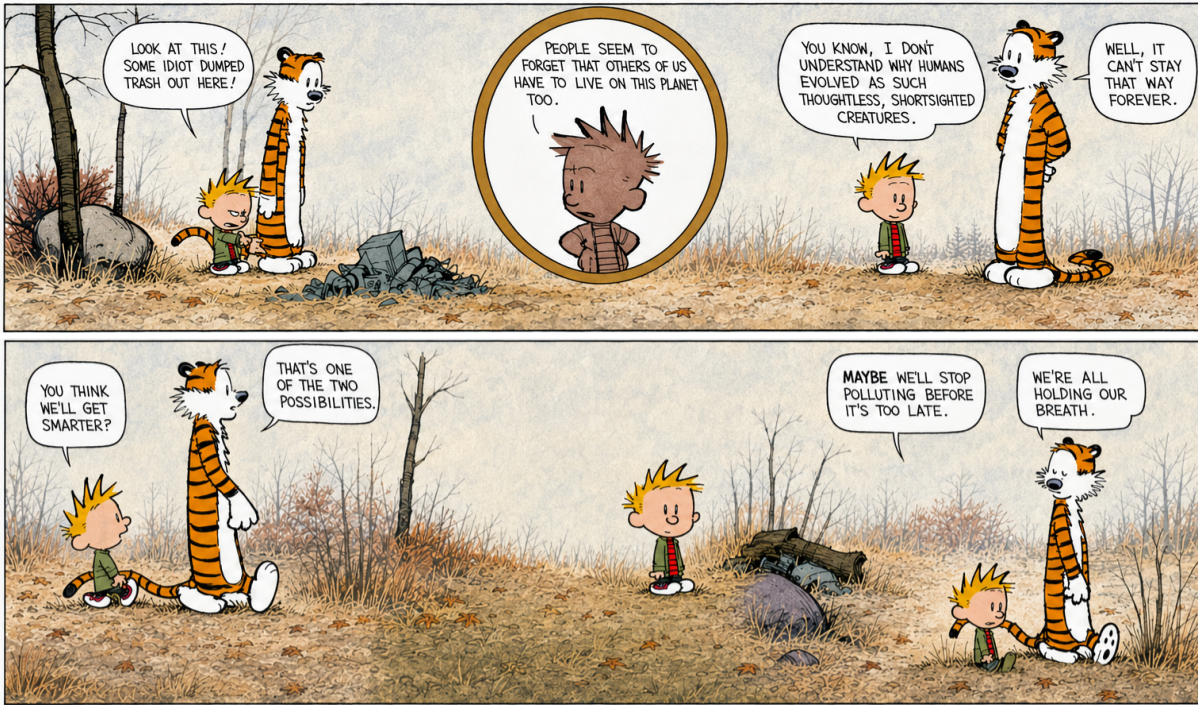
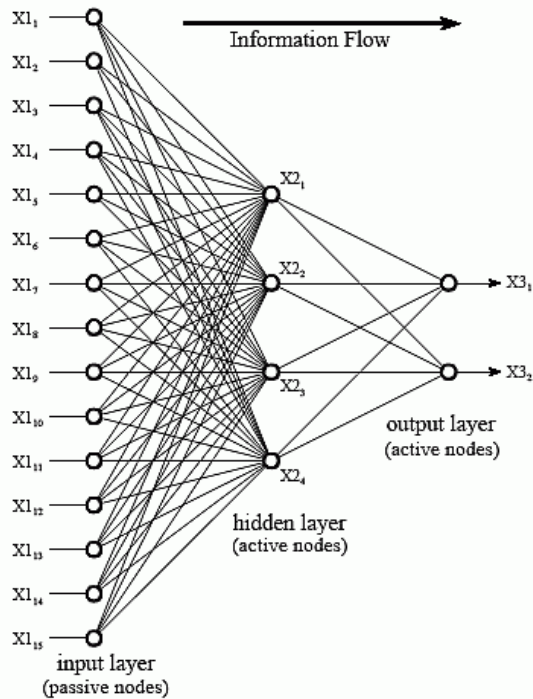


calvin and hobbes by WATSON 10/8/94



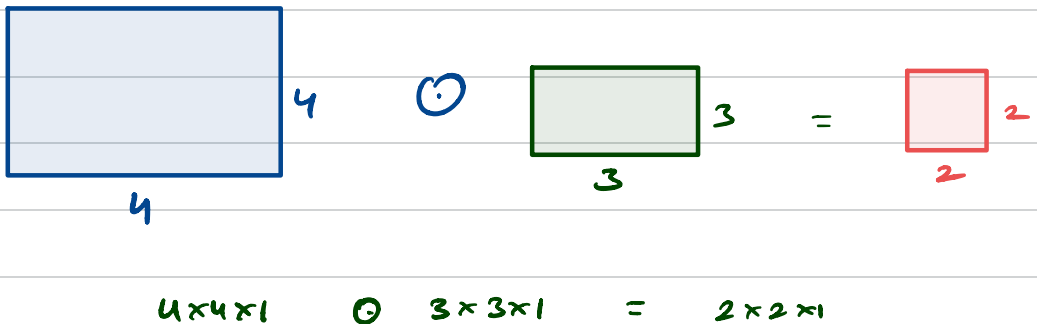
Agenda

FIGURE 26-5
Neural network architecture. This is the most common structure for neural networks: three layers with full interconnection. The input layer nodes are passive, doing nothing but relaying the values from their single input to their multiple outputs. In comparison, the nodes of the hidden and output layers are active, modifying the signals in accordance with Fig. 26-6. The action of this neural network is determined by the weights applied in the hidden and output nodes.



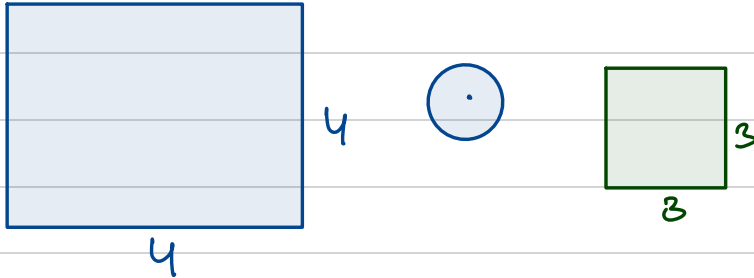
In classical NN the number of neurons decrease as we go deeper into the layer.

But In CNN the number of channels increase as you go deeper into the layers

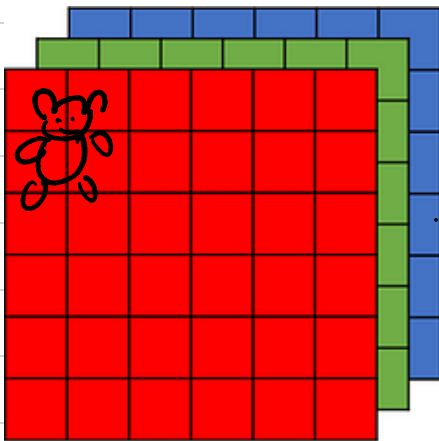


The number of channels in kernel must always match number of channels of image

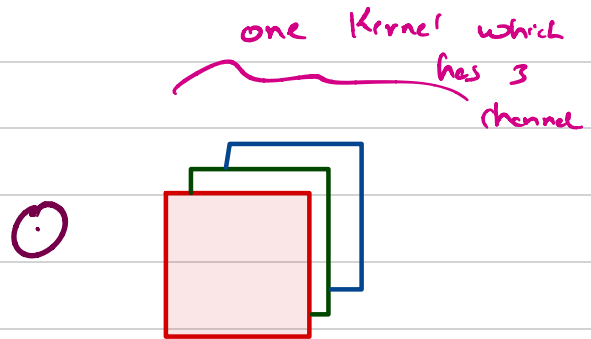
KERNELS WITH MULTIPLE CHANNELS



The number of channels in a kernel must always match the number of channels in the input.



$227 \times 227 \times 3$



$3 \times 3 \times 3$

Output - c1



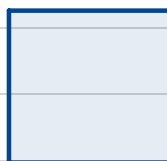
225

Output - c1



225

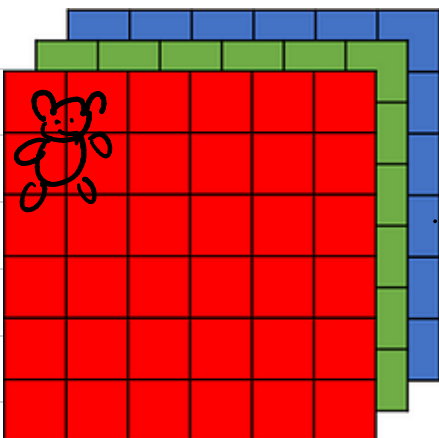
Output - c1



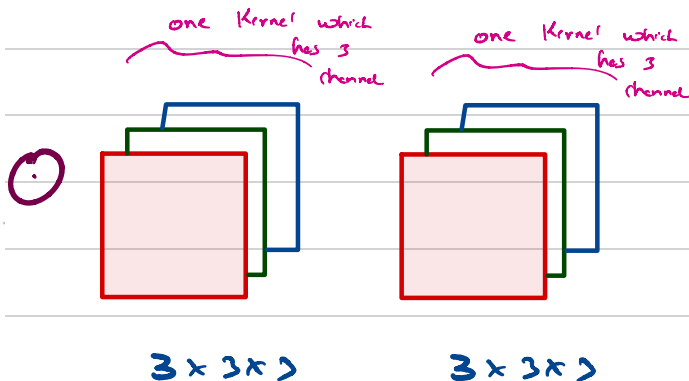
225

$$= 225 \times 225 \times 1$$

[illegible]



$227 \times 227 \times 3$



$$227 \times 227 \times 3 \quad \odot \quad 3 \times 3 \times 3 = 225 \times 225 \times 1$$

$$227 \times 227 \times 3 \quad \odot \quad (3 \times 3 \times 3) \times 2 =$$

$K_1: 3 \times 3 \times 3$ (Dog's Non)

$K_2: 3 \times 3 \times 3$ (Cat's eyes)

$$227 \times 227 \times 3 \quad \odot \quad K_1 = 225 \times 225 \times 1$$

$$227 \times 227 \times 3 \quad \odot \quad K_2 = 225 \times 225 \times 1$$

$$227 \times 227 \times 3 \quad \odot \quad (3 \times 3 \times 3) \times 2 = 225 \times 225 \times 2$$

($225 \times 225 \times 1$) $\times 2$ and $225 \times 225 \times 2$ are same.

INTUITION - MULTI CHANNEL CONU

CASE-I Multi-spectral Imaging in Agriculture.

→ Capture Image data at specific wavelength across Electromagnetic spectrum.

→ Eg. Following spectrum

→ RGB spectrum

* For visual information what humans see

→ Infrared spectrum

* Healthy plants reflect more near infrared

NDVI (Normalized Diff) ^(NIR light)
Vegetation Index

→ Shortwave Infrared (SWIR)

* For water content in soil.

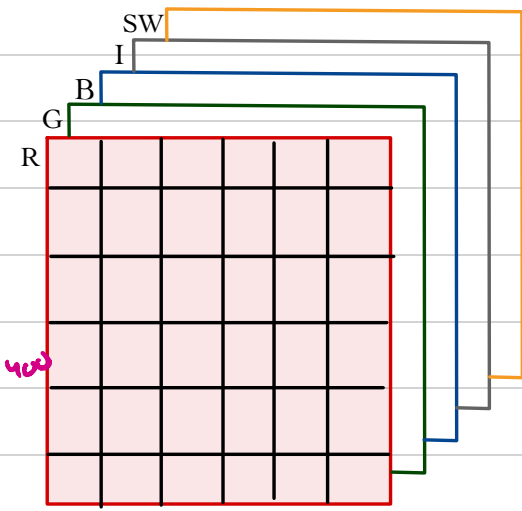
* Water absorbs SWIR signals strongly so it appears darker.

It x w x s

→ Zone Plant
← Hardware

→ For what / kernels might look for.

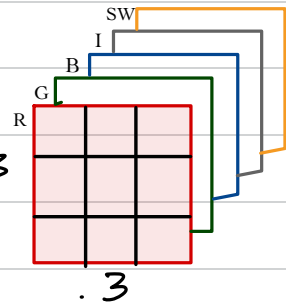
- (i) Signs of Stress
- (ii) Optimizing water use
- (iii) Crop health
- (iv) Yield prediction.



400



3



3

398 x 398 x 1

$$400 \times 400 \times 3 \odot (3 \times 3 \times 5) \times 10 = 398 \times 398 \times 10$$

400x400x3	(3x3x3)x32	398x398x32	RF of 3x3
398x398x32	(3x3x32)x64	396x396x64	RF of 5X5
396x396x64	(3x3x64)x128	394x394x128	RF of 7X7
394x394x128	(3x3x128)x256	392x392x256	RF of 9X9
392x392x256	(3x3x256)x512	390x390x512	RF of 11X11

No of channels in output depends on no of kernels used in previous layer.

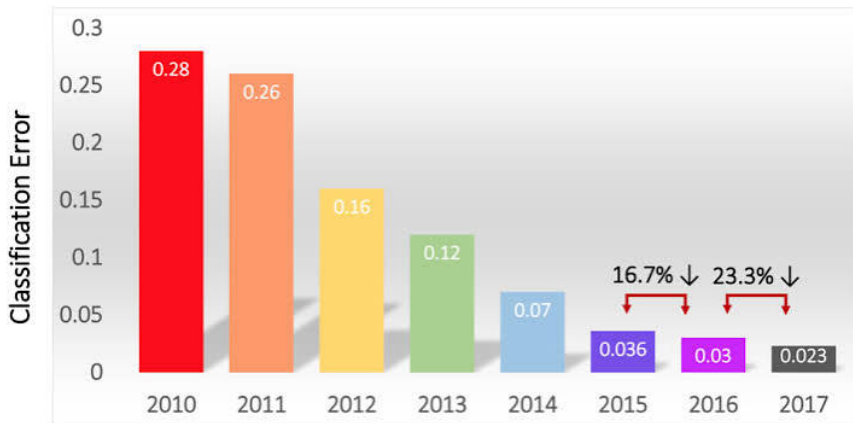
So, in neural network, the number of neurons decreases with subsequent layer. But in convolution, the number of neurons or kernels increase the number of channels across subsequent layers. Why?

Kernels are like weights, and channels like neurons in neural network.

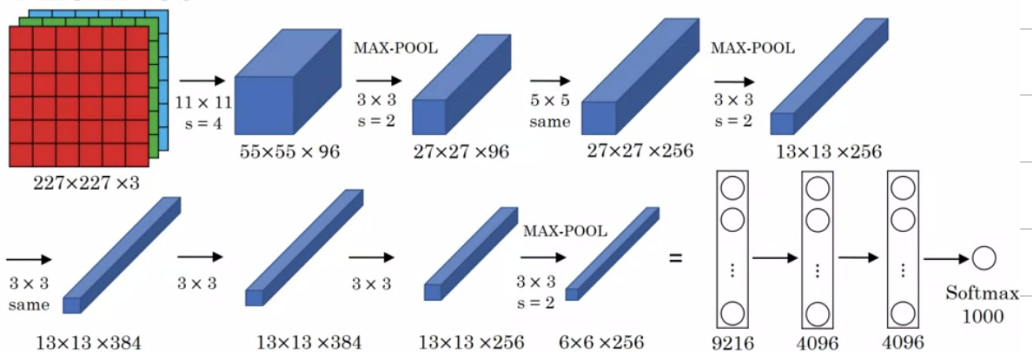
$$400 \times 400 \times 64 \quad \odot \quad (3 \times 3 \times 64) \times 1$$

$$(\text{kernel_height} * \text{kernel_width} * \text{input_channels} + \text{bias}) * 96$$

Classification Results (CLS)



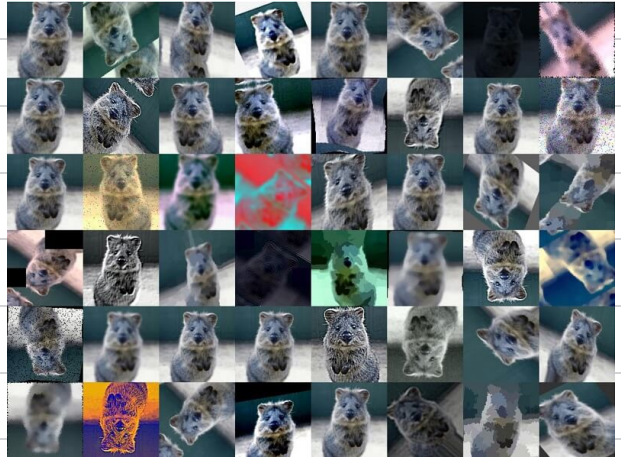
AlexNet



Fancy Augmentation

we've Level Augmentations

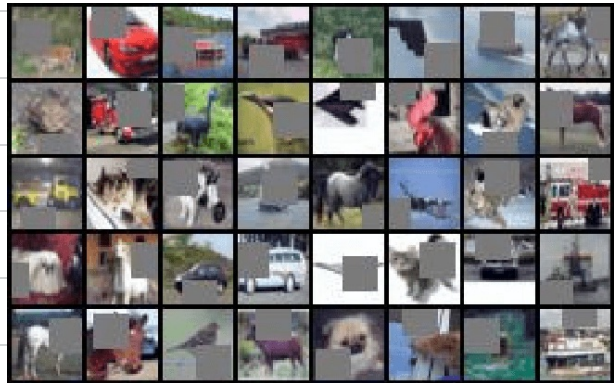
most of it is
built in / keras

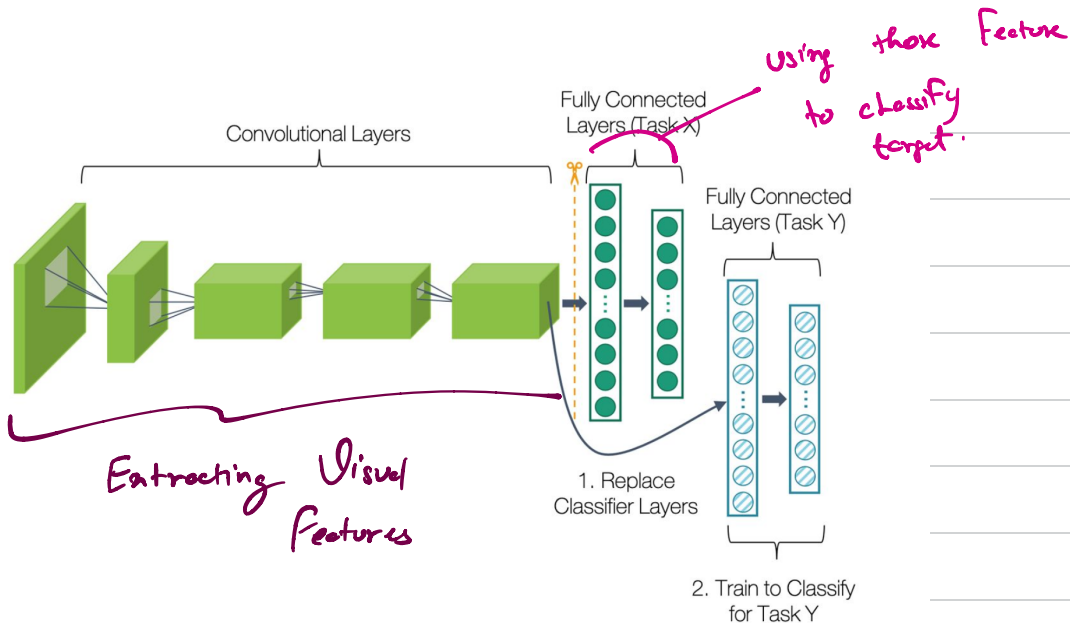


CUT OFF — Higher Levels

V.V. Effective.

Augmentation



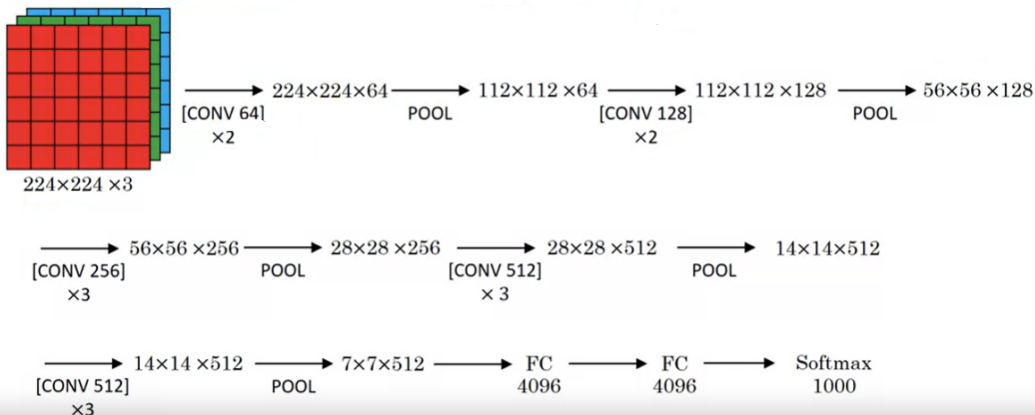


Entirely on Coding, simple CNN, Transfer learning for our own dataset, and augmentation

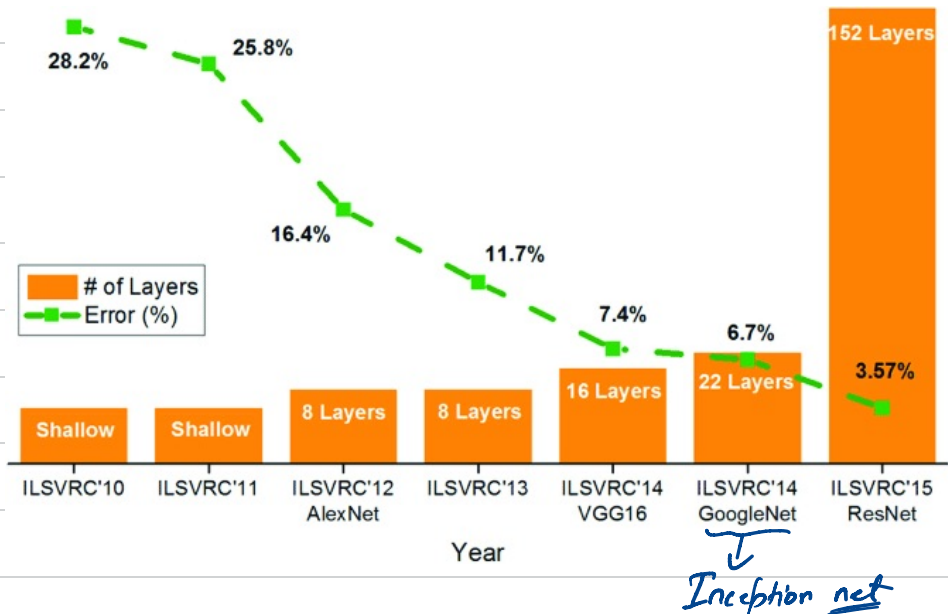
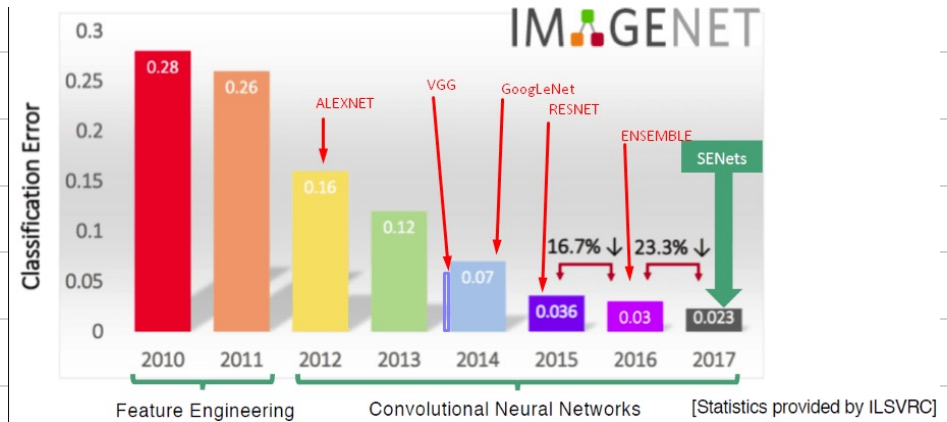
VGG - 16

CONV $\approx$ 3×3 filter, $s = 1$, same

MAX-POOL = 2×2 , $s = 2$

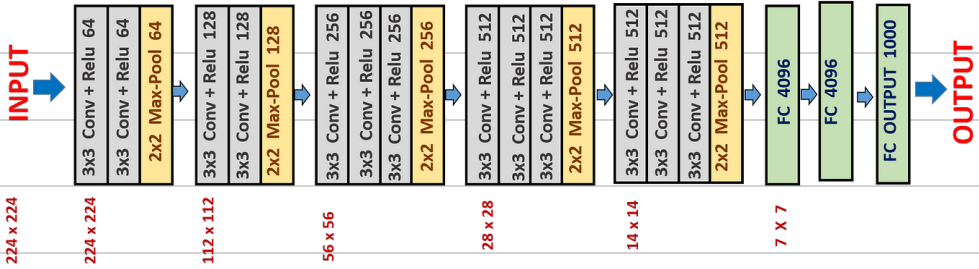


IMAGENET COMPETITION

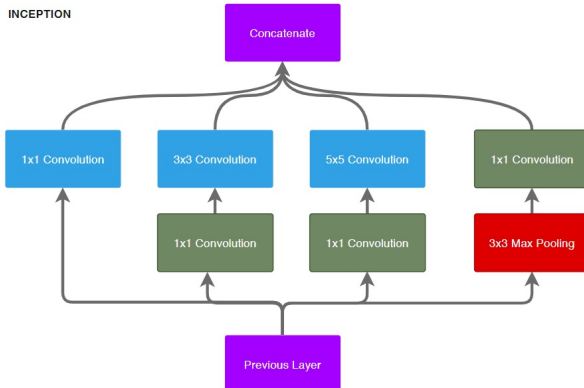
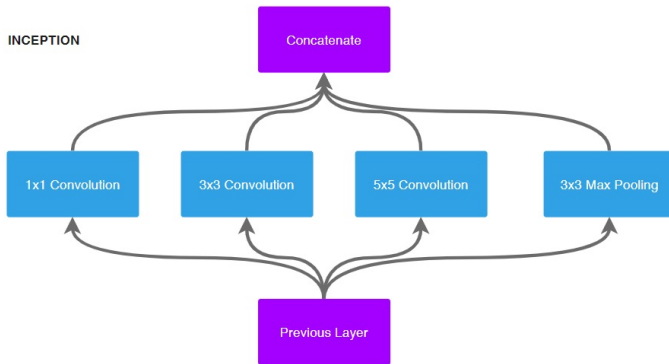


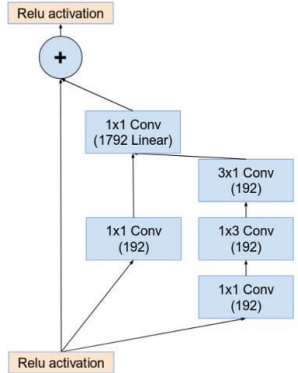
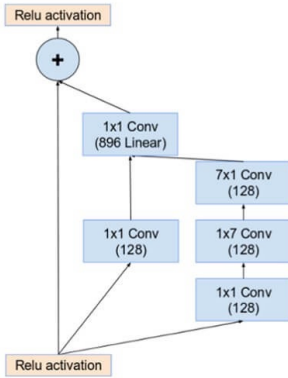
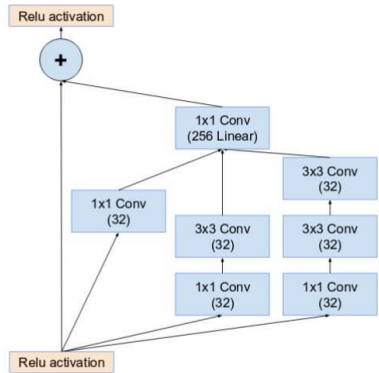
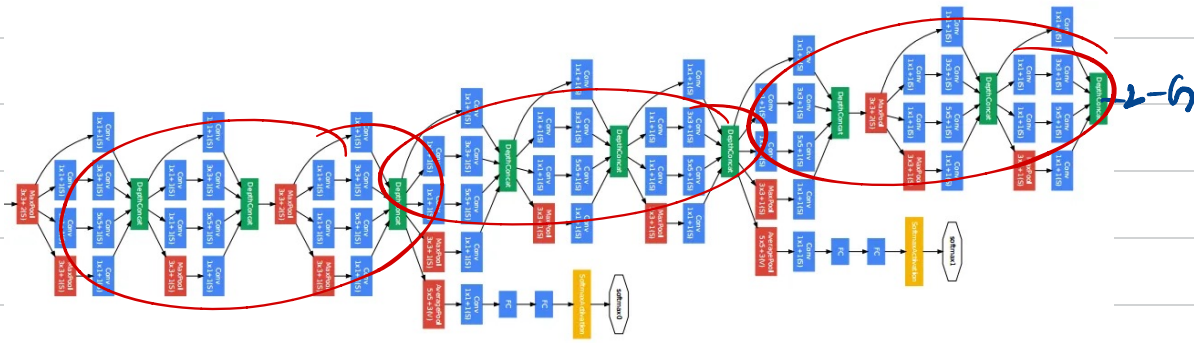
PROBLEMS WITH TRAINING DEEPER MODELS

VGG-16



GOOGLE NET / INCEPTION

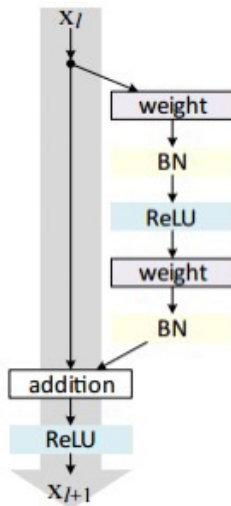




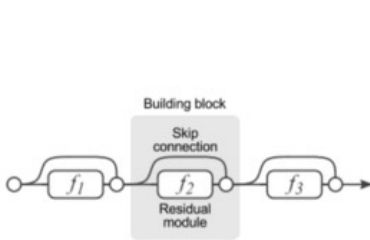
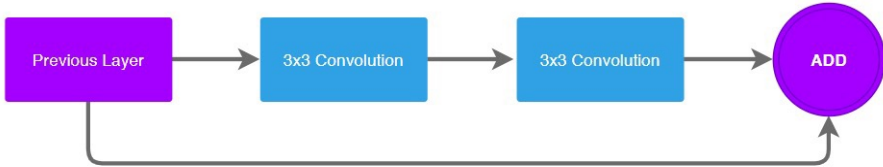
RESIDUAL NET

→ Can I train a CNN model with 1 million Identity layers??

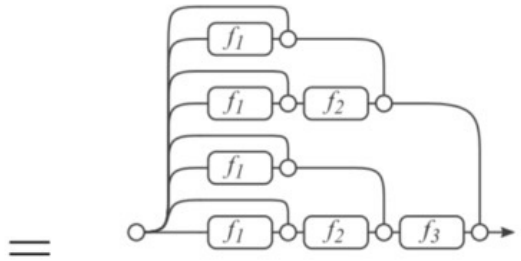
def func(x):
 return x.



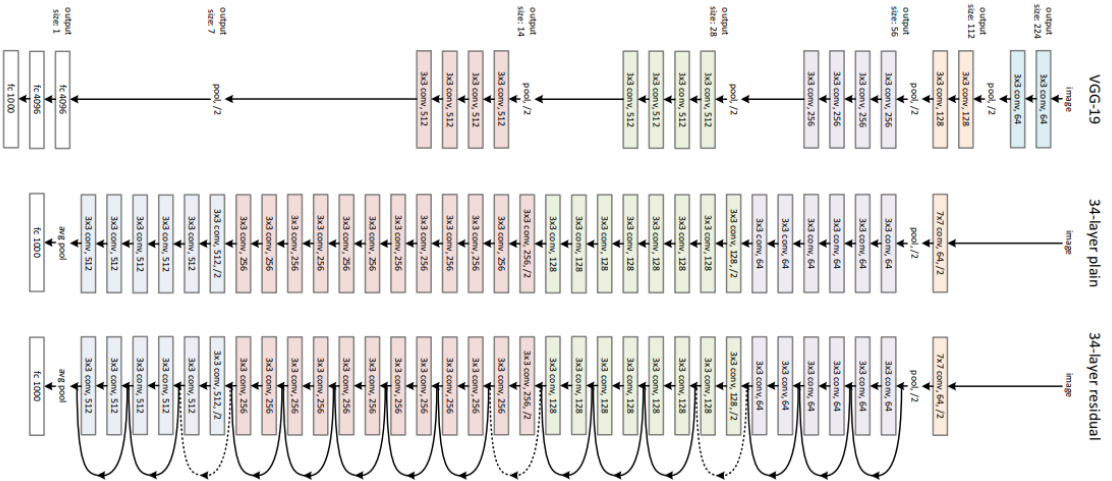
RESNET



(a) Conventional 3-block residual network



(b) Unraveled view of (a)





FRANÇOIS . CHOLLET

He says: They do this because it's what you should be doing. Residual connections with different shapes should be handled via a learned linear transformation between the two tensors, e.g. a 1x1 convolution with appropriate strides and border_mode, or for Dense layers, just matrix multiplication.

CS229 -> CS224

-> CS231

-> CS336